Motivating Future Doctors? Short-Run Evidence from a Large-Scale Survey Experiment in China

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Author note

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Abstract

Physician shortages and attrition from medical training impose costs on healthcare systems, yet little is known about how different forms of encouragement shape career intentions during professional training. We embed a six-arm randomized field experiment in the 2021 China Medical Student Survey (N = 213,326 across 123 institutions). All five treatment messages increase immediate stated intention to practice medicine relative to a control mean of 76.8%. In the adjusted specification, messages emphasizing social respect and an altruistic role model increase intention by 3.0 and 2.8 percentage points, respectively, compared with 1.2 percentage points for physician salary information and 1.1 percentage points for descriptive peer norms. Formal pairwise tests confirm that the social-respect and role-model effects each exceed the salary and peer-information effects after adjustment for multiple comparisons. The Effects of the three professional-encouragement messages decline across baseline professional-motivation quintiles; for social respect, the estimated effects are 5.3 percentage points in the lowest quintile and 0.5 percentage points in the highest. These findings suggest that professional-encouragement messages are most effective at strengthening immediate stated career intentions among students with lower baseline professional motivation.

Keywords: Professional motivation, career intention, medical student, survey experiment, information intervention

JEL classification: J24, D91, I23, O15

1. Introduction

Attrition along the pathway from medical training to medical practice may reduce the returns on public and private investments in education and weaken the supply pipeline for future physicians. It may also signal a mismatch between students' initial program choices and their later career preferences. These issues are economically relevant because medical training is costly, entry into practice is institutionally regulated, and patients cannot fully observe the quality of care (Arrow, 1963). Students' willingness to enter medicine is therefore an important consideration alongside expanding training capacity.

Although the global health workforce has expanded, many countries still face shortages that constrain progress toward universal health coverage (Haakenstad et al., 2022). China has expanded medical education, but this expansion may strain training capacity, and some graduates do not enter medical practice (Hou et al., 2014; Wang, 2021). In a 2019 survey conducted by the National Center for Health Professions Education Development (NCHPED), 14% of graduates from five-year undergraduate medical programs reported that they did not plan to continue practicing medicine (Ye et al., 2022).

In this study, we examined whether brief information messages strengthen medical students' intention to practice medicine, and how responses vary with their baseline professional motivation. We addressed these questions using the 2021 China Medical Student Survey (CMSS), with a sample of 213,326 students across 123 institutions, covering approximately half of the students enrolled in undergraduate medical programs in China. Medical education is a particularly informative setting because entry into physician practice generally requires an undergraduate medical degree followed by licensing and postgraduate training. Before the experimental module, the survey measured students' stated career intentions and baseline attitudes, including their professional motivation. We embedded a six-arm randomized experiment in the 2021 CMSS to estimate the immediate effect of five text-and-graphic messages on stated career intentions. T1 emphasized the social significance of medicine; T2 emphasized the historical and contemporary social recognition of physicians; T3 presented an altruistic physician role model; T4 highlighted physician salaries; and T5 reported peers' intentions to practice medicine. These messages address professional meaning, reputation, identification with role models, financial returns, and descriptive social norms. We compare their effects while recognizing that each message may engage several motivational channels.

All five messages increase the probability that students report intending to become physicians. Relative to the control-group mean of 76.8%, the estimated effects are 1.8 percentage points for T1, 3.0 for T2, 2.8 for T3, 1.2 for T4, and 1.1 for T5 in the preferred specification. Formal cross-treatment comparisons show that the messages emphasizing social respect and an altruistic role model each have larger effects than those providing salary and peer information after Holm adjustment for multiple comparisons.

We next explored how responses to professional encouragement vary with students' baseline professional motivation. Students who already identify strongly with medicine may be more recep-

tive to messages emphasizing professional meaning, recognition, and exemplary practice. Alternatively, these messages may have greater influence among students whose attachment to medicine is weaker, because such messages introduce or reinforce considerations that are less firmly established in their career intentions. Our exploratory heterogeneity analysis complements the average treatment effect estimates by examining how responses vary with baseline professional motivation (Athey and Imbens, 2016). The estimated effects of T1–T3 decline monotonically across baseline motivation quintiles. For T2, the estimated effect is 5.3 percentage points in the lowest quintile and 0.5 percentage points in the highest. These findings suggest that professional encouragement is particularly effective in strengthening immediate stated career intentions among students with lower baseline professional motivation. Because outcomes were elicited shortly after exposure, the survey captures immediate responses related to career intentions, professional attitudes, and intended engagement with training. These reported responses may also reflect social desirability bias or respondents' perceptions of the researchers' expectations. Determining whether these effects persist and influence subsequent career choices requires follow-up data.

Our study contributes to the literature on motivation and career choice by examining responses to encouragement during professional training. Economic theories show how identity, incentives, and mission alignment can shape behavior (Akerlof and Kranton, 2000; Bénabou and Tirole, 2003; Besley and Ghatak, 2005). Much of the empirical literature examines how career opportunities attract individuals with different motivations (Finan et al., 2017). Dal Bó et al. (2013) found that higher wages attracted more qualified applicants without reducing public service motivation in Mexico. Ashraf et al. (2020) found that emphasizing career opportunities changed the applicant pool for health-worker positions in Zambia and that the workers ultimately recruited were more talented and equally prosocial. Deserranno (2019) found that higher expected earnings signaled a lower social contribution and discouraged prosocially motivated candidates from applying for health-promoter jobs in Uganda. We extend this inquiry to students who have already chosen medical education but whose intentions to enter the profession may remain responsive to encouragement.

By testing all five messages within the same population of medical students, we provide direct evidence on their relative effectiveness. Prior experiments have shown that messages can influence career choices: Ajzenman et al. (2024) found that priming altruistic identity and highlighting monetary rewards encouraged teachers in Peru to apply to disadvantaged schools, while Gulzar and Khan (2025) found that publicly delivered prosocial messages encouraged political candidacy in Pakistan. Related experiments among Chinese medical students document responses to prosocial and role-model messages (Ye et al., 2022; Li et al., 2023). Our comparison brings professional significance, social respect, an altruistic role model, salary information, and peer intentions into one design, enabling direct comparison of their effects under common conditions.

This comparison is grounded in research on the different considerations that shape the attractiveness of work. Bénabou and Tirole (2006) distinguish intrinsic, material, and reputational motives. Research on the meaning of work (Cassar and Meier, 2018; Nikolova and Cnossen, 2020), so-

cial respect and reputational rewards (Ellingsen and Johannesson, 2007; Kosfeld and Neckermann, 2011; Gallus and Frey, 2016), and role models (Merton, 1957; Morgenroth et al., 2015; Serra, 2025) informs the design of the professional-encouragement messages. These perspectives guide our comparisons and interpretation, while we recognize that the messages may engage overlapping motives.[1]

We also provide evidence on how responsiveness to professional encouragement varies with existing attachment to medicine. Self-determination theory emphasizes autonomy, competence, and relatedness (Deci and Ryan, 1985, 2000; Ryan and Deci, 2000); related work examines intrinsic and prosocial motivation, task significance, and relational sources of motivation (Grant, 2008a, 2008b; Grant and Shandell, 2022). Motivational interventions have also been studied in education (Lazowski and Hulleman, 2016). Building on these perspectives, we use baseline professional motivation to examine which students respond more strongly to encouragement. The decline in the effects of T1-T3 across baseline motivation quintiles suggests that professional messages may be particularly consequential for students whose attachment to medicine is weaker, providing a basis for studying how encouragement can be tailored to students' initial motivation levels.

These findings are relevant to research on physician supply because they concern career intentions during medical training, a stage before entry into practice, complementing research on international migration by healthcare professionals (Bhargava and Docquier, 2008; Abarcar and Theoharides, 2024). China's medical education system has been criticized for emphasizing technical knowledge while giving insufficient attention to humanities and ethical education (Hou et al., 2014; Wang, 2021). Our findings highlight the relevance of communicating the meaning and social value of medicine during training. They also provide a basis for studying whether sustained professional encouragement can support students' progression into medical practice.

2. Medical education in China and the China Medical Student Survey

Medical education in China starts at the undergraduate level. After high school, students enter medical schools based on their National College Entrance Examination (NCEE) scores. Most students enter a five-year undergraduate program covering general studies, basic medical training, clinical education, and clerkships. Students subsequently apply for graduate study or residency training (Zhu et al., 2016).[2] All medical students must take the National Medical Licensing Examination (NMLE) one year after graduating from undergraduate medical education. Passing the NMLE is required to obtain a physician qualification certificate.

[1] Related research examines motivation during employment. Fracchia et al. (2023) compare honorific awards intended to raise social status with videos intended to increase perceived task significance among volunteer community health workers, finding clearer improvements in household health from the awards.

[2] Two programs provide a continuous training pathway. The "8-year" track prepares prospective physician-scientists, awarding bachelor's and doctorate degrees after eight years of training (Wu et al., 2021). The "5+3" track combines five years of undergraduate medical education with a three-year Master of Medicine (MM) program.

For comparison, entry to medical school in the United States generally requires an undergraduate degree, whereas students in China's standard five-year medical program enter directly after upper secondary schooling. This difference in entry timing helps explain why the study of career intentions during undergraduate medical education is especially relevant in China.

The China Medical Student Survey (CMSS) is an annual survey administered by the National Center for Health Professions Education Development (NCHPED) and first piloted in 2019. NCHPED was established by the Ministry of Education and the National Health Commission to support medical-education reform, develop undergraduate education standards, and implement accreditation in China. For the 2021 survey, NCHPED invited medical schools to participate. Each participating institution appointed a trained program manager, who distributed the online survey through official university channels to eligible students. Respondents completed the survey on a dedicated platform. The CMSS collects self-reported information on demographics, pre-college background, educational experiences, satisfaction, and career plans. The 2021 CMSS was administered in June. No financial or other incentives were offered for participation. Although subsequent CMSS waves were conducted, the present analysis uses the 2021 survey experiment and does not include respondent-level follow-up outcomes linked to the original treatment assignments.

2.1. Study sample

The analysis used the 2021 CMSS and includes 213,326 students from 123 of the 193 institutions that offer undergraduate medical programs (63.7% of those institutions). Students were enrolled in rural five-year, five-year, and eight-year programs. The analysis sample represents approximately half of undergraduate medical students in China. [3]

Participating institutions distributed invitations, and students participated voluntarily. The 63.7% figure refers to institutional coverage. The analysis does not include school-specific counts of eligible students, individual records for nonparticipants, or school-level comparisons with nonparticipating institutions. Appendix Table A1 reports summary statistics and balance results. Random assignment supports comparisons across messages within this study population, and school fixed effects account for school-level differences.

Students were, on average, 21 years old and were distributed broadly across the five undergraduate years. 44.3% were men and 39.3% were only children. Approximately one-quarter reported a father with a college degree, and 22.3% reported a mother with a college degree. The median household income category was RMB 30,000–80,000 per year. 331.8% were classified as having a family medical background. The student-weighted regional shares were 38.3% for eastern China, 28.0% for central China, and 33.7% for western China.

[3] In 2010, the Chinese government introduced a tuition-waived, rural-oriented five-year medical education program to support the supply of medical and health professionals in rural and remote areas of central and western China. This program is also known as the “rural 5-year program”.

2.2. Baseline career intention and correlated factors

In collaboration with NCHPED, we included questions in the 2021 CMSS on students' pre-college, college experiences and baseline stated career intention. We collected these measures before the intervention.

We organized the baseline survey items into four descriptive indices: professional motivation, external/contextual reasons for choosing medicine, admission-related evaluations, and college experience. Each index combines related survey items that describe students' orientation toward medicine or their educational experience. The professional-motivation index combines self-assessed academic preparation, interest in medicine, confidence, perceived professional respect, and the prosocial meaning of medicine. The external/contextual index combines employment prospects, encouragement or pressure from others, and expected access to medical resources. The remaining two indices summarize admission-related evaluations and college experiences. Appendix Table B1 reports the item counts, response scales, and internal consistency estimates; the accompanying text lists the component items and describes the scoring approach.

Each baseline index is standardized to a mean of zero and a standard deviation of one, so its coefficient represents the association with a one-standard-deviation increase in that index. Each index should be interpreted in light of its content and construction. The indices vary in item content, number of items, and response scales; the admission-related index, for example, combines three five-point items with one three-point item. Although standardization expresses all indices in standard-deviation units, comparisons across indices also depend on their content and measurement precision.

Cronbach's alpha was 0.753 for the professional motivation index, 0.735 for the external/contextual reasons index, 0.736 for the admission-related evaluations index, and 0.943 for the college experience index. These statistics summarize the internal consistency of the indices. We interpret each index in terms of its constituent items and use the baseline regressions to describe its association with career intention.

The questionnaire presented items in a fixed order, with professional-motivation and external/contextual items adjacent early in the survey and institutional evaluations later. We interpret the descriptive associations in light of this sequence, since order and fatigue may affect later responses. The coefficients describe associations between the measured domains and career intention; they do not rank the domains by substantive importance.

We measured baseline career intention using the following question:

> Baseline question: "After completing your undergraduate studies, do you wish to become a doctor?"

83.2% of respondents answered "Yes," 13.6% answered "Unsure," and 3.2% answered "No." For the descriptive analysis in Table 1, we define an indicator that equals one for "Yes" and zero for

"Unsure" or "No" and estimate linear probability models relating this indicator to the standardized baseline indices.

In Table 1, column (1), a one-standard-deviation increase in professional motivation is associated with a 0.096 higher probability of answering "Yes," which corresponds to 9.6 percentage points, conditional on the listed covariates and school fixed effects. When all four indices are entered jointly in column (5), the corresponding coefficient is 0.078, or 7.8 percentage points. These estimates describe conditional associations between the professional-motivation index and career intention, while they do not identify the causal effect of changes in motivation.

The baseline and experimental analyses address complementary questions. Table 1 describes how the indices relate to existing career intentions. The randomized comparisons estimate the effects of the five messages, and the heterogeneity analysis examines how those effects vary with students' baseline characteristics. Together, these analyses characterize existing professional attachment and responsiveness to encouragement. The baseline results show how students' reasons for studying medicine and their educational experiences relate to their current intentions to enter medical practice. Students on the medical training pathway express a range of career intentions: about 17% answer "Unsure" or "No" to the baseline career question. The positive association between the professional-motivation index and stated intention makes students' attachment to the profession a relevant target for intervention. The index combines interest, preparation, confidence, perceived respect, and professional meaning, each of which may contribute to that attachment. Figure A1 presents the distributions of baseline responses across quintiles of the four indices.

We therefore asked whether bringing particular reasons for pursuing medicine to students' attention can strengthen their current career intentions. This question connects the descriptive evidence to the five messages examined below: professional significance, social respect, an altruistic role model, salary information, and peers' career intentions. The first three emphasize reasons to value and identify with the profession; the latter two provide information about its economic returns and the career intentions of other medical students.

3. Experimental design

The experimental module was placed near the end of the 2021 CMSS, after baseline covariates and pre-treatment career intention had been measured. With equal probability, the online survey system randomly assigned each eligible respondent to one of five message conditions or a no-message control condition. We randomized respondents before presenting the experimental messages.

Although assignment was random, we examined whether the six groups were comparable on observed baseline characteristics. Appendix Table A1 reports descriptive statistics and pairwise comparisons between each treatment group and the control group, followed by joint tests for 32 baseline characteristics. For each characteristic, we estimate a regression on the five treatment indicators and test whether their coefficients jointly equal zero. Each regression includes an intercept and the five treatment indicators, with standard errors clustered at the school level across the 123

schools. The table reports F-statistics and unadjusted p-values. For 28 of the 32 baseline characteristics, the joint F-tests yield p-values of at least 0.05. Four characteristics have unadjusted p-values below 0.05: the indicator for baseline intention not to become a doctor (p = 0.042), low household income (p = 0.027), NCEE score (p = 0.031), and the central-region indicator (p = 0.012). With a sample of this size, even small differences may be statistically significant, and chance alone can produce statistically significant differences under random assignment when multiple characteristics are examined. In addition, a global joint test that regressed treatment assignment on all baseline characteristics yields an F-statistic of 0.973 (p=0.575), indicating no systematic imbalance. Overall, the pairwise comparisons and joint tests suggest that observed baseline characteristics are broadly balanced across the six groups.

Respondents were assigned to one of five text-and-graphic messages or to the no-message control condition. Appendix C reproduces the intervention materials.

T1: Professional significance. The intervention was adapted from a speech delivered by Dr. Qimin Zhan, a leader in the medical science community. In this speech, he described healthcare professionals as vital national resources, cited the fight against COVID-19 to emphasize the key role of health professionals in society's well-being, and called for a culture that honors and supports healthcare providers.

T2: Social respect. This intervention presented two images depicting a child patient and a doctor bowing to each other, one set during the Qing Dynasty in the nineteenth century and the other in contemporary times. The accompanying text emphasized the respect and care between doctors and patients.

T3: Altruistic role model. This intervention described healthcare workers' contributions during the COVID-19 pandemic. It featured Dr. Zhong Nanshan, a leading physician-scientist during the SARS outbreak in 2003 and the COVID-19 outbreak in 2020 in China, emphasizing his humility and dedication to the profession.

T4: Salary information. This intervention presented a graph based on the China Hospital Salary Survey, showing doctors' average salaries in 2019 by city tier in China. The graph showed a national average salary of 185,000 yuan (44,000 USD at purchasing power parity), equivalent to 4.35 times the average earnings in salaried employment in China, and an annual salary growth of 4%.

T5: Peer career intentions. This intervention presented a graph showing medical undergraduates' intention to become doctors after graduation by gender and program type, with a national average of 87%.

The five messages address complementary considerations relevant to students' career intentions. T1 draws attention to medicine's contribution to society by making salient the significance and public-service meaning of medical work (Deci and Ryan, 1985; Ryan and Deci, 2000; Cassar and Meier, 2018). T2 highlights recognition and the doctor–patient relationship, connecting a medical career with social respect and relational rewards (Grant and Shandell, 2022). T3 presents an altruistic physician as a concrete example of professional commitment and as a model with whom

students may identify (Morgenroth et al., 2015; Serra, 2025). T4 supplies salary information relevant to the economic attractiveness of medical practice, while T5 provides a descriptive norm about peers' stated career intentions. Together, these messages address both professional attachment and other considerations that students may weigh when evaluating a medical career. We classified the treatments by their observable content and estimate the effect of each message bundle, acknowledging that the messages may activate overlapping motivational channels.

Beyond average treatment effects, we examined whether the effects of all five messages varied with baseline professional motivation. For T4 and T5, we also examined how the effects varied with household income and prior beliefs about peers' career intentions, respectively. These analyses focused on characteristics relevant to each message's content; we did not prespecify directional predictions, so the patterns discussed below are exploratory.

Baseline professional motivation captures students' existing orientation toward medicine. The index combines interest in medicine, perceived competence and confidence, and views on the profession's respect and social value (Section 2.2 and Appendix B). We examined treatment effects across quintiles of this index. This dimension is particularly relevant to T1–T3, which emphasize medicine's social contribution, professional recognition, and exemplary practice. Students with stronger professional motivation may find these messages more compelling because they resonate with existing interests and professional identification. Alternatively, students with weaker professional motivation may respond more strongly because the messages highlight reasons for pursuing medicine that are less firmly established in their career intentions. For highly motivated students, the same messages may largely reinforce considerations they already value. The analysis explores which of these patterns emerge and whether responsiveness to salary and peer information also varies with baseline professional motivation.

T4 highlights the economic returns to a medical career.[4] Information indicating that salaries exceed students' expectations could increase the perceived attractiveness of a medical career. We examined whether the effect of T4 varies with household income, considering salary information may be especially consequential for students from less affluent families if financial considerations are more salient to them.[5] This analysis concerns heterogeneity by economic background and household income, not a measure of prior salary expectations.

T5 reports peers' intentions to become doctors. We examined whether responses to this message vary with prior beliefs about peers' career intentions. For students who believe that fewer peers intend to enter medicine than the message indicates, the information may increase the perceived attractiveness of the profession or make a descriptive social norm more compelling. Table 2 examines this possibility by heterogeneity in responses by prior beliefs about peers' intentions.

[4]There is a rich literature on how extrinsic incentives may affect intrinsic motivation or altruistic behavior (Bénabou and Tirole, 2006). A crowding-out effect is found in some contexts (Frey and Oberholzer-Gee, 1997; Mellström and Johannesson, 2008; Deserranno, 2019), but not in others (Lacetera et al., 2012; Ashraf et al., 2020; Goette and Stutzer, 2020).

[5]The average salary shown in T4 was 185,000 yuan, compared with the sample's median household-income category of RMB 30,000–80,000.

Among respondents with an observed primary outcome, each of the six arms accounted for 16.2%–16.9% of the analysis sample, with the control group at 16.8%. The questionnaire contained 34 numbered questions for non-graduating students and two additional employment questions for graduating students. The reported mean time spent on the message page was 12 seconds, and the mean total survey duration was 23 minutes.

Appendix Table A7 reports how the cleaned records for medical students were reduced to the final analysis sample. Of 219,396 cleaned records, 214,673 had a recorded assignment, and 4,723 had no recorded assignment. Among assigned respondents, 213,326 had an observed primary outcome and were included in the analysis. Outcome nonresponse ranged from 0% in the control group to 1.53% in the T4 group. After the message page, or the corresponding survey stage in the control arm, respondents completed the final section on post-graduation plans. The primary outcome was based on the following survey item:

> Post-treatment question: "Please evaluate if the following comment matches you: I want to become a doctor after graduation. Choose one answer from Strongly agree, agree, neutral, disagree, strongly disagree."

The primary outcome equals one for "Strongly agree" or "Agree" and zero for "Neutral," "Disagree," or "Strongly disagree." We refer to this measure as stated career intention. Appendix Table A5 reports the baseline and post-treatment response distributions separately. Table A2 uses an alternative indicator coded as one for "Neutral," "Agree," or "Strongly agree"; Table A6 compares this threshold with the main definition with a threshold based on "Strongly agree" alone.[6] Because we measured career intention immediately after the messages, the estimates capture students' immediate evaluations of a medical career. The design allows us to compare responses to different forms of encouragement under common survey conditions.

We estimated linear probability models by ordinary least squares for the post-treatment stated-career-intention indicator. The main specification is:

$$Y_{is} = \alpha + \sum_{a=1}^{5} \beta_a D_{ais} + X_{is}'\gamma + \mu_s + \varepsilon_{is} \tag{1}$$

Here, i indexes students, s indexes schools, and a indexes the five treatment arms. Y_{is} equals one for agreement or strong agreement with wanting to become a doctor. D_{ais} indicates assignment to arm a, with the no-message control omitted. X_{is} contains the covariates; μ_s denotes school fixed effects. The coefficient β_a is the treatment-versus-control difference in probability units, so a positive value indicates stronger stated intention and $100\beta_a$ expresses the difference in percentage

[6]The implemented outcome question uses a five-point agreement scale, extending the binary "yes" or "no" format specified in the pre-analysis plan (Wu et al., 2022). Early pilots and participant feedback motivated this change: the binary format generated missing responses and varying interpretations of the question. The five-point format accommodates a range of views, and the main binary coding follows the prespecified regression equation. Appendix Tables A2 and A6 describe alternative binary coding.

points. Standard errors are clustered by school. The regressions use the observed-primary-outcome sample described above.

4. Results

4.1. Main results

Table 2 reports the main regression results. Column (1) reports unadjusted estimates. Column (2) adds baseline stated career intention, the other listed covariates, and school fixed effects. The estimates are similar across these specifications. In column (2), the point estimates are 1.8, 3.0, 2.8, 1.2, and 1.1 percentage points for T1, T2, T3, T4, and T5, respectively.

Appendix Table A3 reports formal comparisons among the adjusted treatment coefficients. The table reports all ten pairwise comparisons, pointwise 95% confidence intervals, raw p-values, and Holm-adjusted p-values for that ten-test family. After Holm adjustment, the effects of both T2 and T3 exceed those of both T4 and T5. The effect of T1 exceeds that of T5 at the 5% significance level (adjusted $p = 0.049$); the T1–T4 comparison has an adjusted p-value of 0.129. A separate equal-weighted contrast between the average effects of sT1–T3 and T4–T5 is 1.367 percentage points (pointwise 95% CI 0.995–1.739; unadjusted $p < 0.001$). This summary contrast provides an additional comparison between the two sets of messages and is reported separately from the ten-test Holm family.

In Table 2, columns (3) and (4) examine responses to salary and peer information in more detail. Column (3) relates the T4 effect to household income, while column (4) relates the T5 effect to prior beliefs about peers' career intentions. Both interaction tests yield p-values above 0.05.

Figure 1 reports treatment effects on the probabilities of the five post-treatment response categories, and Appendix Table A5 gives the corresponding counts and percentages. Together, they describe how the response distribution varies across experimental arms.

For T1–T3, higher probabilities of "Strongly agree" accompany lower probabilities of "Neutral" responses and "Agree" responses. This pattern indicates stronger endorsement of a medical career in the post-treatment response distribution and complements the binary outcome. All comparisons are between randomized groups.

Alternative cutoffs show how the estimated responses vary across levels of endorsement. Appendix Table A6 reports positive point estimates for all three thresholds. For the "Strongly agree" threshold, the T1–T3 estimates are 3.945, 4.165, and 3.892 percentage points; for "Neutral" or higher, they are 0.333, 0.392, and 0.230 percentage points. The latter outcome has a control-group mean of 0.972 (Appendix Table A2), which is close to the upper bound. These comparisons show that the estimated response magnitude depends on the threshold used to define career intention.

4.2. Treatment heterogeneity by baseline motivation

We estimated Equation (1) separately within quintiles of the baseline professional-motivation index for all five messages, thereby comparing average treatment effects across groups defined before treatment. This analysis examines whether the messages elicit larger responses among students with stronger or weaker baseline professional motivation.

Figure 2 shows that the estimated effects of T1, T2, and T3 each decline monotonically across the five baseline professional-motivation quintiles. For example, the social-respect message (T2) increases stated career intention by 5.3 percentage points in the lowest quintile, compared with 0.5 percentage points in the highest. Across the three professional-encouragement messages, the largest immediate responses occur among students with weaker initial attachment to medicine.

Panel B in Figure 2 reports pairwise tests of equality of treatment effects across quintiles. For each of T1–T3, the lowest- and highest-quintile effects differ at the 5% level, and several other comparisons between lower and higher quintiles are also significant. The effects of T4 and T5 show no consistent gradients, with all pairwise quintile-comparison p-values above 0.05. These exploratory tests report unadjusted p-values for ten pairwise comparisons within each of the five treatment arms. Differences in statistical significance across arms do not establish differences between their gradients.

The observed gradients indicate that treatment effects are larger among students with lower baseline professional motivation. However, students with less settled intentions may respond more strongly to both the professional content and perceived expectations to express commitment. In addition, the binary outcome leaves less scope for upward movement among students with high initial intention. This pattern is consistent with professional-encouragement messages have greater immediate influence when students' attachment to medicine is less firmly established.

Appendix Figure A2 extends the heterogeneity analysis to the baseline external/contextual index. For T2 and T3, effects are also larger in their lowest than in the highest quintiles, with some between-quintile differences being statistically significant. These patterns suggest that responsiveness varies with several aspects of students' orientation toward medicine. The indices capture distinct but potentially related considerations, and the findings are consistent with professional encouragement engaging overlapping motives. Appendix Table A4 complements these results with exploratory comparisons by gender and family medical background. After the reported adjustments, the gender and family-background comparisons provide no clear evidence of heterogeneous effects.

4.3. Professional attitudes and intended engagement

The exploratory secondary outcomes in Table 3 examine whether the messages also affect students' reported professional attitudes and intended engagement with training. These outcomes help characterize responses beyond career intention and assess whether professional encouragement is accompanied by more positive evaluations of medicine, stronger professional identification, and

greater confidence. We examine these responses using eleven outcomes measured in the final survey section after exposure to the messages. For each outcome, we estimate Equation (1) with the same controls and school fixed effects as in Table 2, using standard errors clustered at the school level. Appendix B provides the item wording and explains how to interpret the items.

We organized these outcomes according to the aspects of students' professional orientation they capture. Greater appreciation of professional meaning should appear in students' evaluation of medicine as a profession, awareness of doctors' responsibilities, and their view that medicine is more than a means of earning a living (columns 7, 9, and 10). Stronger professional identification and recognition should be reflected in students' pride in being a medical student and their perceived respect for doctors (columns 8 and 11). Greater confidence and aspiration should be reflected in students' confidence about becoming an excellent doctor and optimism about the healthcare industry (columns 5 and 6). T1–T3 emphasize different aspects of professional encouragement, each with implications for several attitudes. We therefore assess response coherence across the relevant outcomes.

T1–T3 have positive, individually significant effects on all seven professional-attitude outcomes in columns 5–11, using the unadjusted tests reported in Table 3. For example, the professional-significance, social-respect, and role-model messages increase reported professional pride by 3.1, 4.3, and 4.1 percentage points, respectively, and increase confidence in becoming an excellent doctor by 2.3, 3.1, and 2.8 percentage points. All three also increase reported professional value, responsibility, and respect.

The career-intention and study-plan outcomes connect these attitudinal responses to students' stated aspirations. T1–T3 increase agreement with the two alternative expressions of career intention by 2.4–3.8 percentage points (columns 1–2). They also increase reported plans to participate in academic and practical activities by 2.8–4.0 percentage points and plan to follow developments in the profession by 3.3–4.6 percentage points (columns 3–4). These results indicated that the messages also affect students' reported plans for engagement with medical training.

The salary and peer-information messages also increase several reported professional attitudes and study plans. Each increase reported professional pride by 1.4 percentage points. This pattern is consistent with several motivational channels making medicine more attractive: information about economic returns or peers' intentions can also influence professional attitudes. T1–T3 generate a broader pattern of positive estimates across these outcomes, consistent with their direct emphasis on professional meaning, recognition, and exemplary practice.

Together, these results show that the messages affect students' immediate reports of professional attitudes, career intentions, and intended engagement with training. Because these outcomes were elicited at the same time, they describe accompanying responses and may share social-desirability effects. They do not identify causal mediation or explain why treatment effects vary with baseline professional motivation. The reported plans provide a starting point for follow-up research on engagement with training. Table 3 reports 55 unadjusted treatment-versus-control tests across eleven

outcomes. The pattern across outcomes is exploratory; these tests do not compare the treatment coefficients with one another.

5. Discussion and conclusion

This large-scale survey experiment shows that information messages strengthen medical students' intention to practice medicine immediately after exposure. All five messages have positive estimated effects relative to the control group. Formal comparisons show that messages emphasizing social respect and an altruistic role model have larger effects than messages providing salary and peer information. The three professional-encouragement messages generate their largest effects among students with lower baseline professional motivation. These findings identify which forms of encouragement elicit stronger career intentions and which students respond most strongly within the study population.

The heterogeneity results show that professional encouragement has diminishing effects as attachment to medicine becomes more established. The accompanying improvements in reported attitudes provide further evidence consistent with this interpretation. These findings highlight the potential value of emphasizing professional meaning, recognition, and exemplary practice for students whose attachment to medicine is still developing. However, the evidence concerns immediate self-reports among participating students with observed outcomes. When interpreting these findings, it is important to take into account voluntary participation, between-group differences in outcome nonresponse, possible reporting effects, and the binary nature of the outcome measure.

Brief professional messages delivered through an online survey provide a practical starting point for studying encouragement in medical education. Where feasible, linking the original treatment assignments to subsequent academic effort, clinical placements, graduation, licensing, and employment would establish whether immediate changes in stated intention are accompanied by later changes in training and career outcomes. Such evidence could inform the design and timing of sustained encouragement.

Conflict of interest statement

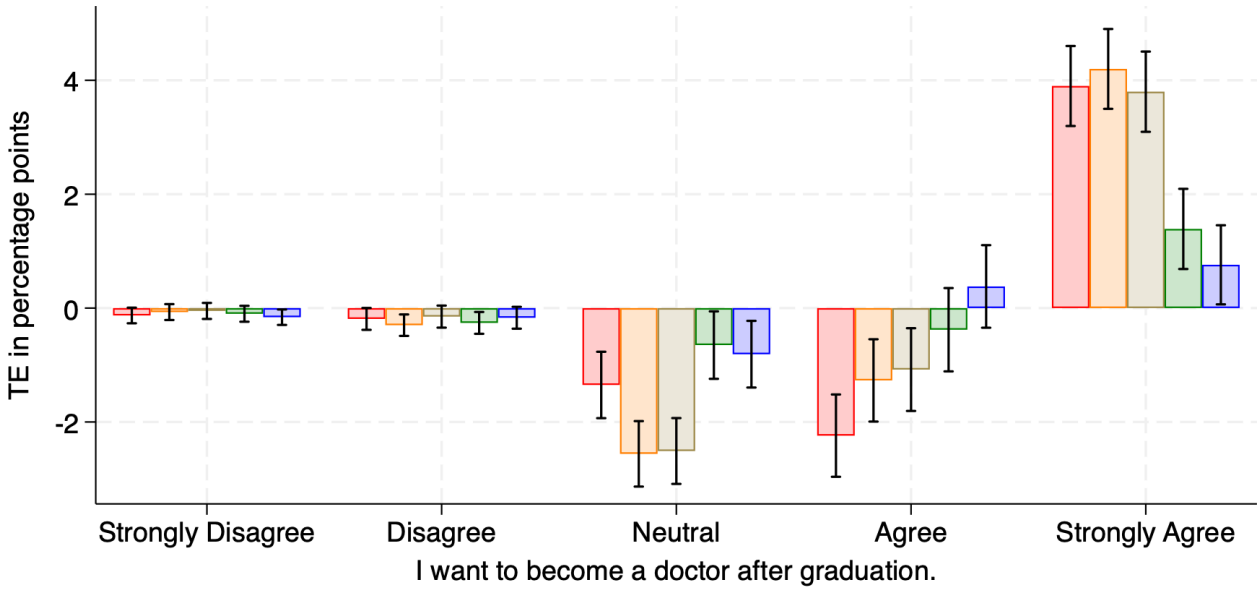
We declare no competing interests.

Data availability statement

Access to the study data is subject to approval by the participating centers and a signed data-access agreement. Requests should be directed to the corresponding author.

Figures and tables

Figure 1. Treatment effects on post-treatment career-intention categories

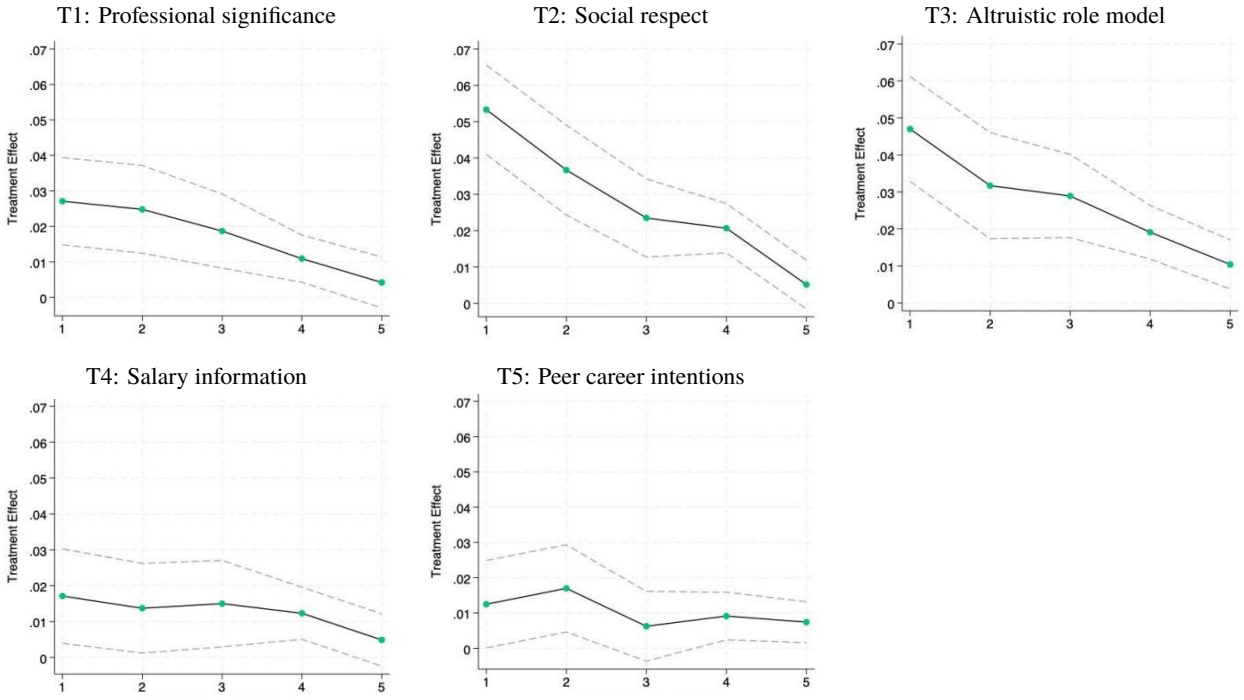


T1: Professional significance T2: Social respect
T3: Altruistic role model T4: Salary information
T5: Peer career intentions

Notes: Bars show treatment-versus-control differences in the probability of each response category, expressed in percentage points. Error bars show 95% confidence intervals. Table A5 reports the response counts and percentages.

Figure 2. Treatment heterogeneity by baseline professional motivation

Panel A. Treatment effects by professional-motivation quintile



Horizontal axes: baseline professional-motivation quintile (1 = lowest; 5 = highest). Vertical axes: treatment effects in probability units.

Panel B. Unadjusted p-values for equality of effects between quintiles

Quintiles	T1	T2	T3	T4	T5
Q1–Q2	0.801	0.049	0.142	0.710	0.656
Q1–Q3	0.346	0.001	0.082	0.809	0.432
Q2–Q3	0.493	0.127	0.773	0.893	0.211
Q1–Q4	0.027	<0.001	<0.001	0.535	0.611
Q2–Q4	0.046	0.024	0.123	0.861	0.265
Q3–Q4	0.182	0.671	0.183	0.722	0.617
Q1–Q5	0.002	<0.001	<0.001	0.103	0.460
Q2–Q5	0.004	<0.001	0.010	0.259	0.148
Q3–Q5	0.022	0.002	0.002	0.148	0.842
Q4–Q5	0.168	0.002	0.076	0.108	0.684

Notes: Panel A plots estimates by baseline-index quintile. Panel B reports the ten pairwise comparisons within each treatment arm. All comparisons in Panel B use unadjusted p-values.

Table 1. Predictors of baseline intention to practice medicine

Baseline index	(1)	(2)	(3)	(4)	(5)
Professional motivation	0.096***				0.078***
	(0.002)				(0.001)
External/contextual reasons		0.034***			−0.007***
		(0.001)			(0.001)
Admission-related evaluations			0.081***		0.043***
			(0.002)		(0.002)
College experience				0.057***	0.019***
				(0.002)	(0.001)
Other controls	Yes	Yes	Yes	Yes	Yes
School fixed effects	Yes	Yes	Yes	Yes	Yes
R^2	0.159	0.110	0.135	0.123	0.170
Observations	213,326	213,326	213,326	213,326	213,326

Notes: OLS linear probability models. The dependent variable equals one for "Yes" and zero for "Unsure" or "No" at baseline; its mean is 0.832. The four indices are standardized to a mean of zero and a standard deviation of one. Coefficients are in probability units; multiply by 100 for percentage points. Columns (1)–(4) include one index at a time; column (5) includes all four. Coefficients describe cross-sectional associations with baseline career intention. All columns include the other covariates and school fixed effects. Parentheses contain school-clustered standard errors. Stars indicate unadjusted p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2. Effects on post-treatment career intention

Variable	(1)	(2)	(3)	(4)
T1: Professional significance	0.017***	0.018***	0.018***	0.017***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T2: Social respect	0.029***	0.030***	0.030***	0.030***
	(0.004)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T3: Altruistic role model	0.027***	0.028***	0.028***	0.028***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
T4: Salary information	0.010***	0.012***	0.014***	0.012***
	(0.004)	(0.003)	(0.004)	(0.003)
	[0.008]	[0.001]		
T5: Peer career intentions	0.011***	0.011***	0.011***	0.010***
	(0.003)	(0.003)	(0.003)	(0.003)
	[0.001]	[0.001]		
Low family income			−0.011***	
			(0.002)	
Low family income × T4			−0.002	
			(0.004)	
Overestimate dropout				−0.037***
				(0.003)
Overestimate dropout × T5				0.004
				(0.005)
Baseline career intention	—	0.555***	0.555***	0.551***
		(0.005)	(0.005)	(0.005)
Control mean	0.768	0.768	0.768	0.768
Other controls	No	Yes	Yes	Yes
School fixed effects	No	Yes	Yes	Yes
R^2	0.001	0.306	0.306	0.307
Observations	213,326	213,326	213,326	213,326

Notes: Standard errors in parentheses are clustered at the school level. The MHT-adjusted p-values are reported in square brackets. Multiple-hypothesis testing follows List et al. (2019). Controls include gender, ethnicity, political affiliation, only-child status, grade, enrolled program, family medical background, senior high school type, ideal career, parental expectations, career training participation, paternal and maternal ISEI (International Socio-Economic Index) levels, parental education, family income, NCEE (National College Entrance Examination) score, and college admission expectations. Coefficients are in probability units; multiply by 100 for percentage points. Stars denote unadjusted p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Appendix Table A3 uses a separate Holm adjustment for ten cross-treatment comparisons.

Table 3. Secondary professional attitudes and study plans

Outcome	(1) Career resolve	(2) Ideal career	(3) Planned activities	(4) Staying informed	(5) Professional confidence	(6) Sector outlook	(7) View of the profession	(8) Professional pride	(9) Sense of duty	(10) Professional meaning	(11) Social respect
T1	0.024***	0.025***	0.028***	0.033***	0.023***	0.027***	0.032***	0.031***	0.011***	0.014***	0.026***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)
T2	0.038***	0.032***	0.040***	0.046***	0.031***	0.025***	0.034***	0.043***	0.033***	0.029***	0.028***
	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)
T3	0.038***	0.030***	0.035***	0.041***	0.028***	0.027***	0.033***	0.041***	0.030***	0.022***	0.030***
	(0.004)	(0.003)	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)
T4	0.007***	0.013***	0.013***	0.025***	0.012***	0.018***	−0.003	0.014***	−0.002	0.019***	0.017***
	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)
T5	0.002	0.009***	0.013***	0.021***	0.008**	0.006*	−0.005	0.014***	0.005*	0.004	0.010***
	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)	(0.004)	(0.003)	(0.004)	(0.003)
Control mean	0.723	0.679	0.742	0.685	0.698	0.537	0.500	0.697	0.838	0.417	0.606
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.186	0.250	0.112	0.123	0.172	0.089	0.092	0.128	0.065	0.054	0.081
Observations	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326	213,326

Notes: Columns (1)–(2) concern alternative expressions of career intention, (3)–(4) study plans, and (5)–(11) professional attitudes. Table B2 reproduces the item wording and interpretation. These outcomes were elicited after treatment in the same survey. Controls are as in Table 2; all columns include school fixed effects. Coefficients are in probability units; multiply by 100 for percentage points. Parentheses contain school-clustered standard errors. Stars denote unadjusted p-values: * p < 0.10, ** p < 0.05, *** p < 0.01. The 55 treatment-versus-control tests are exploratory and unadjusted for multiple testing.

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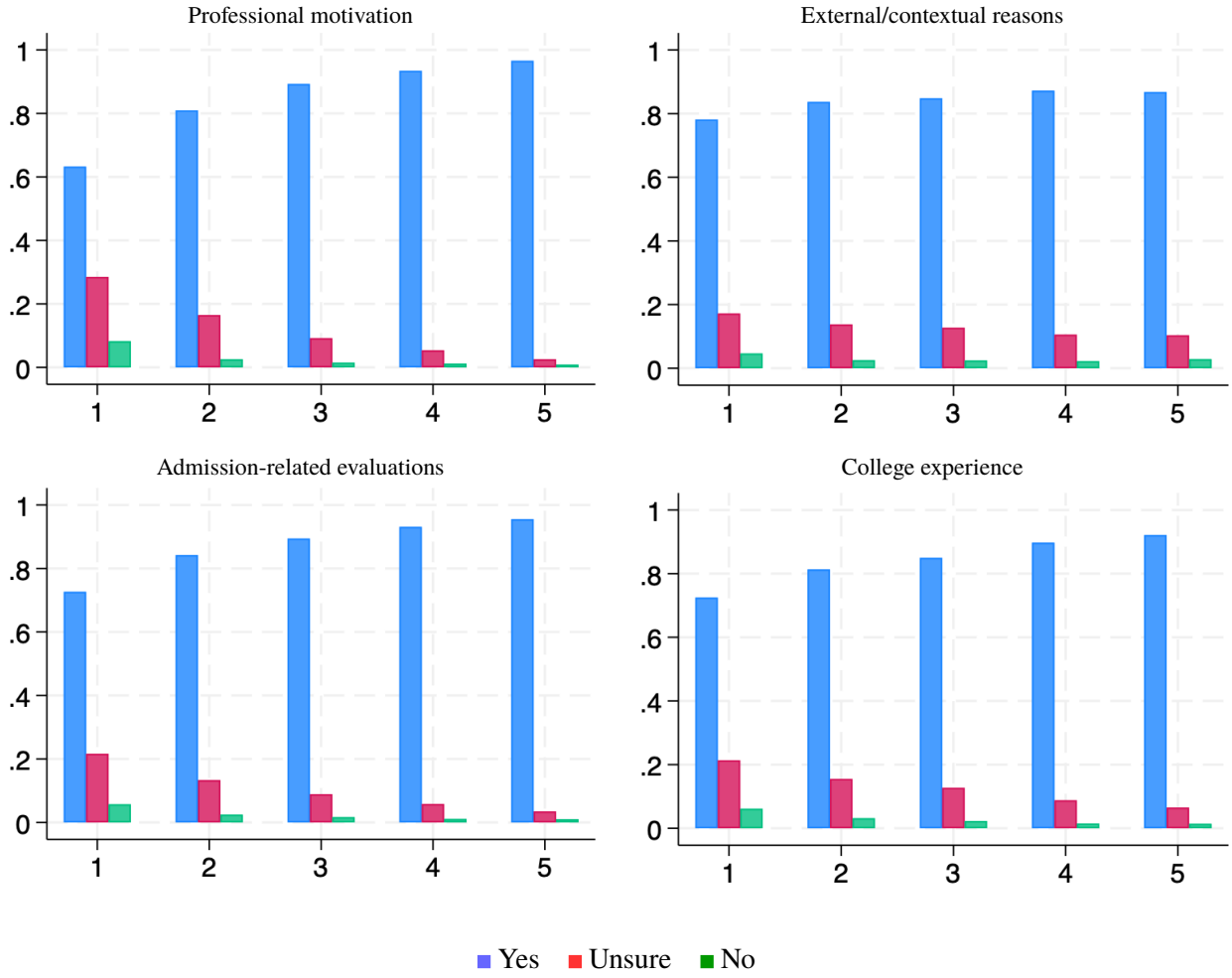
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Appendix A. Additional exhibits

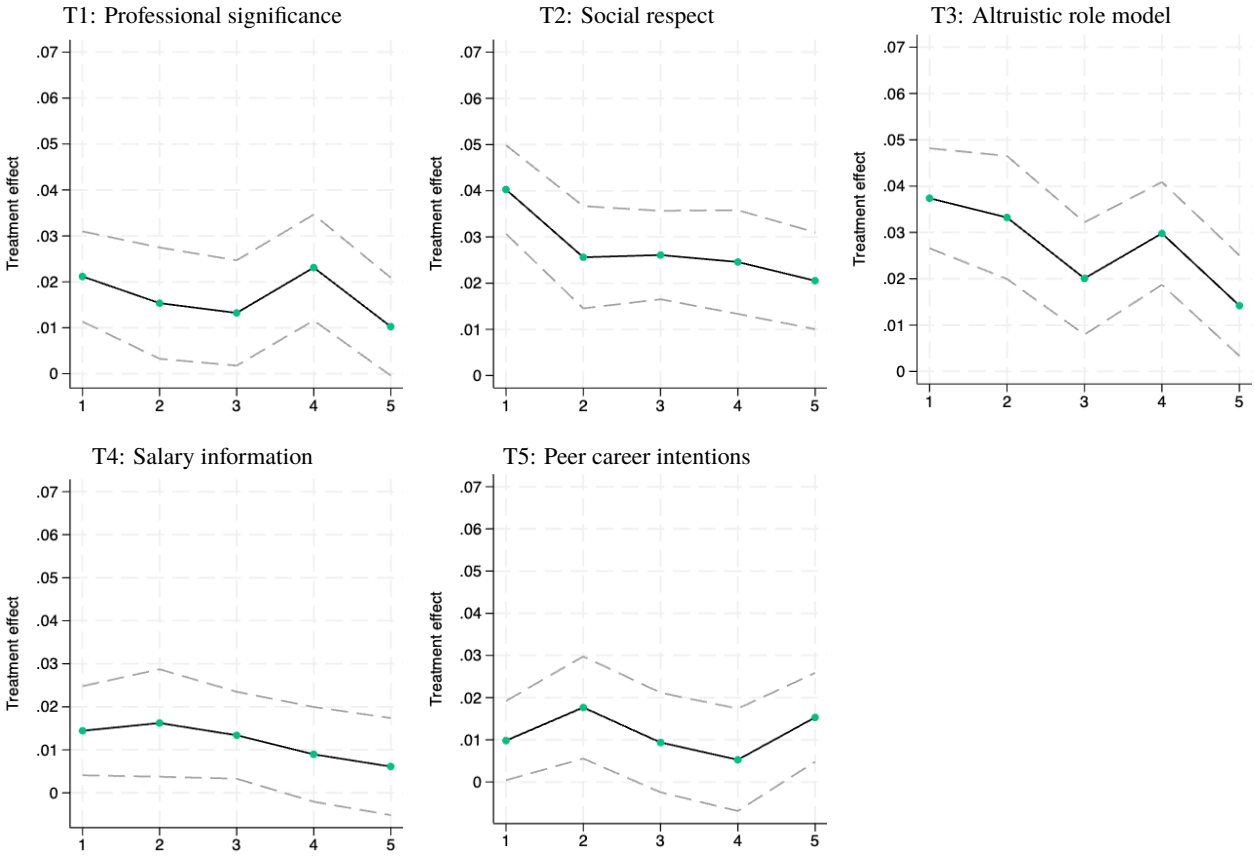
Figure A1. Baseline career intention by baseline-index quintile



Notes: Horizontal axes show quintiles of the corresponding baseline index (1 = lowest; 5 = highest). Vertical axes show response proportions from 0 to 1. The categories refer to the baseline question about becoming a doctor.

Figure A2. Treatment heterogeneity by baseline external/contextual reasons

Panel A. Treatment effects by external/contextual-index quintile



Horizontal axes: baseline external/contextual-index quintile (1 = lowest; 5 = highest). Vertical axes: treatment effects in probability units.

Panel B. Unadjusted p-values for equality of effects between quintiles

Quintiles	T1	T2	T3	T4	T5
Q1–Q2	0.513	0.046	0.649	0.818	0.298
Q1–Q3	0.270	0.034	0.019	0.884	0.954
Q2–Q3	0.807	0.947	0.171	0.743	0.359
Q1–Q4	0.804	0.025	0.333	0.449	0.572
Q2–Q4	0.382	0.896	0.713	0.397	0.167
Q3–Q4	0.237	0.832	0.250	0.580	0.632
Q1–Q5	0.168	0.003	0.003	0.274	0.449
Q2–Q5	0.500	0.476	0.020	0.197	0.756
Q3–Q5	0.716	0.395	0.509	0.367	0.482
Q4–Q5	0.068	0.556	0.027	0.720	0.186

Notes: Panel A plots estimates by baseline-index quintile. Panel B reports the ten pairwise comparisons within each treatment arm. The 50 comparisons are exploratory and use unadjusted p-values.

Table A1. Summary statistics and treatment–control balance

Characteristic	All	Control	T1 – C	T2 – C	T3 – C	T4 – C	T5 – C
Baseline: doctor	0.832	0.833	-0.002	-0.002	-0.002	-0.003	0.001
	[0.374]	[0.373]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Baseline: unsure	0.136	0.135	0.003	0.000	0.002	0.001	-0.001
	[0.343]	[0.342]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Baseline: no doctor intention	0.032	0.031	-0.001	0.002	0.001	0.002	0.000
	[0.176]	[0.175]	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Gender (Male = 1)	0.443	0.441	-0.000	0.001	0.001	0.000	0.007*
	[0.497]	[0.497]	(0.004)	(0.004)	(0.004)	(0.004)	(0.003)
Ethnicity (Han ethnic = 1)	0.885	0.886	-0.002	-0.002	-0.001	-0.000	-0.001
	[0.319]	[0.317]	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
CPC members	0.098	0.098	-0.001	-0.002	0.002	0.001	0.001
	[0.297]	[0.297]	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Only child	0.393	0.397	-0.006**	-0.000	-0.002	-0.008**	-0.006*
	[0.488]	[0.489]	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)
First-year	0.247	0.245	-0.001	0.005	0.002	0.000	0.004
	[0.431]	[0.430]	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Second-year	0.202	0.204	0.002	-0.001	-0.004	-0.006*	-0.002
	[0.402]	[0.403]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Third-year	0.223	0.222	0.002	0.000	-0.000	0.004	0.003
	[0.417]	[0.416]	(0.004)	(0.004)	(0.003)	(0.003)	(0.003)
Fourth-year	0.147	0.147	0.000	-0.002	0.002	-0.001	-0.003
	[0.354]	[0.354]	(0.003)	(0.003)	(0.003)	(0.002)	(0.003)
Fifth-year	0.181	0.181	-0.003	-0.002	-0.001	0.002	-0.001
	[0.385]	[0.385]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Rural 5-year program	0.068	0.069	0.000	-0.003	-0.001	0.000	0.001
	[0.252]	[0.253]	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
5-year program	0.860	0.859	0.003	0.002	0.002	0.001	0.001
	[0.347]	[0.348]	(0.003)	(0.002)	(0.003)	(0.002)	(0.002)
8-year program	0.072	0.072	-0.003	0.000	-0.000	-0.001	-0.001
	[0.258]	[0.259]	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Father's ISEI level	0.639	0.638	0.002	-0.003	0.002	0.005	0.002
	[0.480]	[0.481]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Mother's ISEI level	0.658	0.658	-0.001	-0.005	-0.002	0.004	0.002
	[0.474]	[0.474]	(0.004)	(0.003)	(0.003)	(0.003)	(0.004)
Father's education	0.274	0.276	-0.001	0.003	-0.003	-0.004	-0.003
	[0.446]	[0.447]	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)
Mother's education	0.223	0.223	0.001	0.002	-0.002	-0.002	-0.002
	[0.416]	[0.416]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Low family income	0.644	0.644	0.003	-0.005*	-0.005	0.001	0.003
	[0.479]	[0.479]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
NCEE score	549.987	550.189	-1.001**	0.121	0.578	-0.458	-0.453
	[61.272]	[61.311]	(0.438)	(0.433)	(0.461)	(0.455)	(0.444)
Family medical background	0.318	0.315	0.004	0.004	0.005	0.002	0.001
	[0.466]	[0.464]	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)
Higher admission expectation	0.363	0.363	0.001	-0.003	-0.001	0.002	-0.001
	[0.481]	[0.481]	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)

Characteristic	All	Control	T1 – C	T2 – C	T3 – C	T4 – C	T5 – C
Senior high school type	0.481	0.483	-0.003	-0.002	-0.002	-0.004	-0.002
	[0.500]	[0.500]	(0.003)	(0.004)	(0.004)	(0.004)	(0.004)
Ideal career (1 = doctor)	0.534	0.534	0.001	0.001	-0.003	-0.001	-0.001
	[0.499]	[0.499]	(0.004)	(0.003)	(0.003)	(0.004)	(0.003)
Higher parents' expectation	0.147	0.148	-0.001	-0.004	0.000	0.002	-0.002
	[0.354]	[0.355]	(0.003)	(0.003)	(0.003)	(0.003)	(0.002)
Schools' location (eastern)	0.383	0.386	-0.006*	-0.001	-0.000	-0.004	-0.003
	[0.486]	[0.487]	(0.003)	(0.004)	(0.003)	(0.003)	(0.003)
Schools' location (central)	0.280	0.277	0.009***	0.001	0.004	0.006**	-0.001
	[0.449]	[0.447]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Schools' location (western)	0.337	0.337	-0.003	0.000	-0.004	-0.002	0.004
	[0.473]	[0.473]	(0.003)	(0.003)	(0.004)	(0.003)	(0.003)
University attribution	0.697	0.702	-0.004	-0.005	-0.005	-0.008***	-0.005
	[0.460]	[0.458]	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
University level	0.141	0.141	-0.001	0.003	0.004	-0.002	-0.001
	[0.348]	[0.348]	(0.002)	(0.002)	(0.003)	(0.002)	(0.003)
Career training	0.293	0.295	-0.006**	-0.001	-0.002	-0.002	-0.004
	[0.455]	[0.456]	(0.003)	(0.003)	(0.004)	(0.003)	(0.004)
Observations	213,326	35,753	35,777	36,076	35,301	34,461	35,958

Notes: Baseline (career intention) is represented by three indicators for "Yes," "Unsure," and "No" responses to the pre-treatment question on intention to become a doctor. Male, Han ethnicity, CPC membership, and only-child status equal one for male students, students of Han ethnicity, members of the Communist Party of China, and only children, respectively, and zero otherwise. First- through fifth-year indicators identify the student's year of study. Program indicators distinguish rural-oriented five-year programs, regular five-year programs, and integrated 5+3 or eight-year programs. Father's ISEI level and Mother's ISEI level equal one if the corresponding parent's ISEI score is at most 29 and zero if it exceeds 29. Paternal and maternal higher-education indicators equal one if the corresponding parent received higher education and zero otherwise. Low household income equals one for annual household income of RMB 80,000 or less in 2020 and zero otherwise. The NCEE score is rescaled to a 750-point scale according to the province where the student took the examination. Family medical background equals one if a parent or another family member studied medicine or works in a medical or health institution and zero otherwise. Higher admission expectation equals one if the student's university admission outcome was below their expectations and zero otherwise. Senior high school type equals one for attendance at a key senior high school and zero otherwise. Ideal career equals one if becoming a doctor was the student's ideal career in high school and zero otherwise. Higher parents' educational expectations equal one if parental expectations for the student's educational attainment exceed the student's own expectations and zero otherwise. Eastern, central, and western indicators identify the institution's location. University attribution equals one for independent medical colleges and zero for comprehensive universities. University level equals one for Double First-Class universities and zero otherwise. Career training equals one if the student received career-planning education in high school. "All" and "Control" report means, with standard deviations in square brackets. T1 − C through T5 − C report treatment–control comparisons, with standard errors in parentheses. C denotes the control group. Stars indicate unadjusted p-values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The following panel reports joint balance tests and variable-specific sample sizes.

Table A1 (continued). Joint tests of baseline characteristics

Baseline characteristic	Joint F	p-value
Baseline: doctor	0.366	0.871
Baseline: unsure	0.496	0.779
Baseline: no doctor intention	2.382	0.042
Gender (Male = 1)	0.984	0.431
Ethnicity (Han ethnic = 1)	0.403	0.846
CPC members	0.817	0.540
Only child	2.065	0.074
First-year	0.802	0.551
Second-year	1.768	0.124
Third-year	0.613	0.690
Fourth-year	1.868	0.105
Fifth-year	0.773	0.571
Rural 5-year program	0.892	0.489
5-year program	0.389	0.855
8-year program	0.763	0.578
Father's ISEI level	1.013	0.413
Mother's ISEI level	1.920	0.096
Father's education	1.474	0.203
Mother's education	0.777	0.568
Low family income	2.621	0.027
NCEE score	2.560	0.031
Family medical background	0.697	0.627
Higher admission expectation	0.519	0.762
Senior high school type	0.337	0.889
Ideal career (1 = doctor)	0.591	0.707
Higher parents' expectation	1.064	0.384
Schools' location (eastern)	1.325	0.258
Schools' location (central)	3.070	0.012
Schools' location (western)	0.951	0.451
University attribution	1.497	0.196
University level	1.834	0.111
Career training	0.906	0.480

Notes: Each row reports a regression of one baseline characteristic on all five treatment indicators, with the control group omitted. The F-test assesses whether the five coefficients jointly equal zero. The specification includes an intercept and the five treatment indicators; standard errors are clustered at the school level across 123 schools. All p-values are unadjusted, N=213,326.Given the large number of tests, a few statistically significant differences are expected by chance. A global joint test regressing treatment assignment on all baseline characteristics yields an F-statistic of 0.973 (p= 0.575), indicating no systematic imbalance overall.

Table A2. Alternative outcome: neutral or stronger agreement

Variable	(1)	(2)	(3)	(4)
T1: Professional significance	0.003**	0.003***	0.003***	0.003***
	(0.001)	(0.001)	(0.001)	(0.001)
T2: Social respect	0.004***	0.004***	0.004***	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)
T3: Altruistic role model	0.002	0.002*	0.002*	0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
T4: Salary information	0.004**	0.004***	0.006***	0.004***
	(0.001)	(0.001)	(0.002)	(0.001)
T5: Peer career intentions	0.003***	0.003***	0.003***	0.002**
	(0.001)	(0.001)	(0.001)	(0.001)
Low family income			0.003***	
			(0.001)	
Low family income × T4			−0.004*	
			(0.002)	
Overestimate dropout				−0.015***
				(0.001)
Overestimate dropout × T5				0.004
				(0.003)
Baseline stated career intention		0.102***	0.102***	0.100***
		(0.002)	(0.002)	(0.002)
Control mean	0.972	0.972	0.972	0.972
Other controls	No	Yes	Yes	Yes
School fixed effects	No	Yes	Yes	Yes
R^2	0.000	0.066	0.066	0.068
Observations	213,326	213,326	213,326	213,326

Notes: The outcome equals one for "Neutral," "Agree," or "Strongly agree," and zero for "Disagree" or "Strongly disagree." Coefficients are in probability units; multiply by 100 for percentage points. Controls are as in Table 2. Parentheses contain school-clustered standard errors. Stars indicate unadjusted p-values: * p < 0.10, ** p < 0.05, *** p < 0.01. Appendix Table A6 compares point estimates for the main definition, the neutral-or-higher threshold, and strong agreement alone. Table A2 reports the specifications for the neutral-or-higher threshold.

Table A3. Formal comparisons between treatment coefficients

Comparison	Difference, pp	Pointwise 95% CI	Raw p	Holm p
T1 – T2	−1.218	[−1.696, −0.741]	<0.001	<0.001
T1 – T3	−1.071	[−1.475, −0.667]	<0.001	<0.001
T1 – T4	0.551	[0.018, 1.083]	0.043	0.129
T1 – T5	0.656	[0.145, 1.167]	0.012	0.049
T2 – T3	0.147	[−0.340, 0.634]	0.550	1.000
T2 – T4	1.769	[1.160, 2.378]	<0.001	<0.001
T2 – T5	1.875	[1.330, 2.420]	<0.001	<0.001
T3 – T4	1.622	[1.115, 2.128]	<0.001	<0.001
T3 – T5	1.727	[1.183, 2.271]	<0.001	<0.001
T4 – T5	0.106	[−0.480, 0.692]	0.721	1.000
Mean (T1, T2, T3) −Mean(T4, T5)	1.367	[0.995, 1.739]	<0.001	<0.001s

Notes: Differences are expressed in percentage points from the fully adjusted Table 2 specification, with school-clustered standard errors. Confidence intervals are pointwise. Holm p-values adjust the ten pairwise comparisons only. The equal-weighted summary contrast is separate and has an unadjusted p-value.

Table A4. Exploratory heterogeneity by gender and family medical background

Panel A. Gender

Arm	Women pp	Men pp	Men – women pp	Pointwise 95% CI	Raw p	Holm p
T1	1.759	1.759	0.000	[−1.106, 1.107]	0.999	1.000
T2	2.510	3.568	1.058	[0.028, 2.088]	0.044	0.398
T3	2.316	3.480	1.163	[0.130, 2.197]	0.028	0.276
T4	1.039	1.424	0.385	[−0.699, 1.468]	0.483	1.000
T5	0.989	1.250	0.260	[−0.704, 1.225]	0.594	1.000

Panel B. Family medical background

Arm	Absent pp	Present pp	Present – absent pp	Pointwise 95% CI	Raw p	Holm p
T1	1.832	1.598	−0.234	[−1.370, 0.902]	0.684	1.000
T2	3.180	2.539	−0.642	[−1.719, 0.436]	0.241	1.000
T3	2.897	2.682	−0.215	[−1.315, 0.885]	0.699	1.000
T4	1.286	1.039	−0.247	[−1.582, 1.088]	0.715	1.000
T5	1.328	0.612	−0.716	[−1.918, 0.486]	0.241	1.000

Notes: Effects and differences are expressed in percentage points. The interaction models use Table 2 controls, school fixed effects, and school-clustered standard errors. The Holm adjustment covers all ten individual interactions across Panels A and B. Confidence intervals for differences are pointwise. The joint test of five gender interactions has raw p = 0.027 and Holm p = 0.053 across the two moderator-level joint tests. The family-background joint test has raw p = 0.665. Gender interactions compare responses to the same intervention materials across male and female students. Family medical background denotes the survey indicator used in the analysis.

Table A5. Response distributions

Panel A. Baseline

Response	All	Control	T1	T2	T3	T4	T5
Yes	177,441	29,786	29,749	29,992	29,323	28,612	29,979
	(83.18)	(83.31)	(83.15)	(83.14)	(83.07)	(83.03)	(83.37)
Unsure	29,073	4,843	4,953	4,890	4,840	4,700	4,847
	(13.63)	(13.55)	(13.84)	(13.55)	(13.71)	(13.64)	(13.48)
No	6,812	1,124	1,075	1,194	1,138	1,149	1,132
	(3.19)	(3.14)	(3.00)	(3.31)	(3.22)	(3.33)	(3.15)
Valid answers	213,326	35,753	35,777	36,076	35,301	34,461	35,958

Panel B. Post-treatment

Response	All	Control	T1	T2	T3	T4	T5
Strongly disagree	1,847	341	294	318	317	292	285
	(0.87)	(0.95)	(0.82)	(0.88)	(0.90)	(0.85)	(0.79)
Disagree	3,545	657	590	554	598	545	601
	(1.66)	(1.84)	(1.65)	(1.54)	(1.69)	(1.58)	(1.67)
Neutral	40,669	7,287	6,810	6,429	6,307	6,799	7,037
	(19.06)	(20.38)	(19.03)	(17.82)	(17.87)	(19.73)	(19.57)
Agree	89,944	15,348	14,559	15,031	14,772	14,662	15,572
	(42.16)	(42.93)	(40.69)	(41.66)	(41.85)	(42.55)	(43.31)
Strongly agree	77,321	12,120	13,524	13,744	13,307	12,163	12,463
	(36.25)	(33.90)	(37.80)	(38.10)	(37.70)	(35.29)	(34.66)
Valid answers	213,326	35,753	35,777	36,076	35,301	34,461	35,958

Notes: Cells report n (%), with percentages calculated within columns. Both panels describe the 213,326-observation sample with an observed primary outcome. Table A7 reports outcome availability among assigned respondents. The panels summarize the baseline and post-treatment questions separately, preserving their respective wording and response scales.

Table A6. Point estimates under three outcome thresholds

Arm	Agree or strongly agree	Neutral or higher	Strongly agree only
T1	1.759	0.333	3.945
T2	2.977	0.392	4.165
T3	2.830	0.230	3.892
T4	1.208	0.389	1.577
T5	1.102	0.321	0.748

Notes: Point estimates from the full-control specification, expressed in percentage points. Each threshold equals one for the named categories and zero otherwise. The table compares estimated magnitudes across outcome definitions; inferential statistics for the main and neutral-or-higher definitions appear in Tables 2 and A2, respectively.

Table A7. Recorded assignments and primary-outcome availability

Arm	Recorded assignment	Analyzed	Missing %
Control	35,753	35,753	0.00
T1	35,926	35,777	0.41
T2	36,272	36,076	0.54
T3	35,475	35,301	0.49
T4	34,996	34,461	1.53
T5	36,251	35,958	0.81
Total	214,673	213,326	0.63

Notes: Counts show recorded assignment and primary-outcome availability for the 219,396 cleaned medical students' records, including 4,723 records without a recorded assignment. "Analyzed" denotes respondents with an observed primary outcome. Missing percentages use recorded assignments as denominators.

Appendix B. Baseline indices and secondary outcomes

The four baseline indices summarize different aspects of students' orientation toward medicine and their educational experiences. Table B1 provides an overview of their composition. Each index is constructed by summing its component items and standardizing the resulting score to mean zero and standard deviation one; the item wording and scales are detailed below.

Table B1. Baseline index composition and internal consistency

Baseline index	Items	Response scales	Alpha
Professional motivation	5	Five-point agreement	0.753
External/contextual reasons	3	Five-point agreement	0735
Admission-related evaluations	4	Three five-point items and one three-point item	0.736
College experience	11	Five-point satisfaction; 1 overall medical-education satisfaction and 10 institutional-experience items	0.943

Notes: Alpha denotes reported Cronbach's alpha and summarizes internal consistency. Each composite is expressed in standard-deviation units and interpreted according to its item content and response scales. The admission-related index combines items with different response ranges. Standardization does not equalize measurement precision across the indices.

Professional motivation index

The professional-motivation index sums responses to five statements:

1. Achieved good grades in subjects related to this major during high school.
2. Interested in medical-related majors.
3. Confident in achieving success in this field.
4. Believe that being a doctor is respected.
5. Medicine is a sacred and noble profession dedicated to saving lives.

Each statement uses a five-point agreement scale: 1 = strongly disagree, 2 = disagree, 3 = neutral, 4 = agree, and 5 = strongly agree. The raw score is the sum of the five items and is standardized for the regressions. The first item captures students' broad self-assessment of preparation in subjects related to their chosen major. Together, the items describe preparation, interest, confidence, perceived respect, and professional meaning.

External and contextual reasons index

This index combines employment prospects, social influence over the choice of major, and anticipated access to medical resources. We use the label external/contextual reasons because these items include both expected benefits and circumstances surrounding the choice of medicine. The index sums five-point agreement responses to three statements:

1. Studying medicine offers good employment opportunities.

2. My parents, relatives, or teachers encouraged (or required) me to choose a medical major.

3. Studying medicine can provide convenience for me, my family, and friends to access medical resources in the future.

Admission-related evaluations index

The raw score sums four items. The first three use five-point scales and the fourth uses a three-point scale. The standardized index retains this combination of response ranges:

1. Please evaluate your satisfaction with your major admission. (1 = least satisfied; 5 = most satisfied.)

2. Before studying your current major, how well did you understand this major? (1 = did not understand at all; 5 = understood very well.)

3. When choosing a university major, what did you think about the working environment of medical professionals in our country? (1 = very bad, 5 = very good)

4. Based on your actual situation, estimate the gap between your university admission results and your expected goals. (1 = worse than expectation, 2 = as expected, 3 = better than expectation).

College experience index

This index sums responses on five-point satisfaction scales across seven curriculum items and ten institutional-experience items:

1. Overall satisfaction with medical education: "I am satisfied with the medical education I have experienced at my institution." (1 item).

2. Are you satisfied with the other aspects of the institution, including Facilities, Electronic resources supporting student self-study, Elective course system, Grading system, Award and recognition system, Student representation in relevant school committees, Academic guidance and counseling, Psychological counseling services, Career planning and guidance, Clubs and extracurricular activities. (10 items in total.)

Table B2. Secondary outcomes and their interpretive roles

Table 3 column	Statement	Interpretive role
1	Medical education has made me more determined to become a doctor.	Alternative expression of career intention
2	Being a doctor is my ideal career choice.	Alternative expression of career intention
3	I will actively participate in academic and practical activities related to my field.	Intended participation
4	I will keep up with and follow the latest developments in my professional field.	Intended engagement
5	I am confident that I can become an excellent doctor.	Confidence
6	I am optimistic about the future prospects of the healthcare industry.	Professional outlook
7	The profession of a doctor is one of the best professions in today's society.	Evaluation of the profession
8	I feel proud to be a medical student when communicating with students from other majors.	Professional identification
9	I am aware of the responsibilities that a doctor should undertake.	Perceived professional responsibility
10	Being a doctor is more than just a means of making a living.	Professional meaning
11	Being a doctor is highly respected.	Perceived social recognition

Notes: These items were elicited after treatment on five-point agreement scales. They describe accompanying professional attitudes, career aspirations, and study plans. Responses may reflect overlapping motivational channels and shared reporting effects. Because the items were measured at the same time, they describe accompanying responses and do not identify causal mediation.

Appendix C. Intervention materials

Treatment 1. Professional significance

Original version

医务人员是国家最宝贵的资源

詹启敏

（中国工程院院士、北京大学常务副校长、医学部主任、全国医学教育发展中心主任）



一场突如其来的疫情突袭了全球，给各国人民健康和经济发展带来了巨大的创伤，也显示了医学和医学教育在国家安全体系中的重要地位，医学和医学教育是提供全民健康和保证。

医务人员是国家最宝贵的资源。中国自古以来便有“大医精诚”的理念。我们需要进一步凝聚社会共识：医务人员是国家最宝贵的资源，是守卫人民生命安全和健康的卫士，应提倡全社会尊医重卫、崇尚医疗健康行业，形成共建健康中国的良好氛围。

English translation

Healthcare professionals are the nation's most valuable resource

Qimin Zhan

(Academician of the Chinese Academy of Engineering, Executive Vice President of Peking University, Dean of the Faculty of Medicine, Director of the National Centre for Health Professions Education Development)



An unexpected pandemic struck the world, causing significant harm to the health and economic development of people across various countries. It also underscored the importance of medicine and medical education in the national security system, as they are essential for ensuring the health of the population.

Healthcare professionals are the nation's most valuable resource. Since ancient times, China has upheld the concept of “Great Physicians Possess Sincerity and Integrity (Da Yi Jing Cheng).” We need to further solidify social consensus: healthcare professionals are the country's most valuable resource and the guardians of people's life and health. It is crucial to advocate for respect towards the medical profession across society, to esteem the healthcare industry, and to foster a favorable environment for building a healthy China together.

Treatment 2. Social respect

Original version

医者仁心，救人性命，无量功德。一组跨越百年的相似照片，真切体现了医者与患者之间的尊重和关爱。



English translation

Doctors possess benevolence, saving lives with boundless merit. A series of similar photos spanning over a century vividly reflects the respect and care between doctors and patients.



Treatment 3. Altruistic role model

Original version

自从 COVID-19 以来，医生与护士和其他在前线工作的医护人员一起，成为了“最美逆行者”、名副其实的英雄——他们冒着自己的生命危险为他人服务！谁言医者皆文弱，不破疫情誓不回！全国各地千千万万的医生，在本次疫情中体现出作为医者的担当，就像钟南山院士所说，**他相信他们都是英雄！**



《人民日报》曾这样评价钟南山：既有国士的担当，又有战士的勇猛。他回应得最多的一句话是“我不过是一个看病的大夫”。穿上这身白衣，便选择全力以赴。**以钟南山为榜样，全国已有上万名年轻医务工作者奔赴湖北。**他们恪尽职守、一丝不苟，只因医者仁心，不辱使命。

English translation

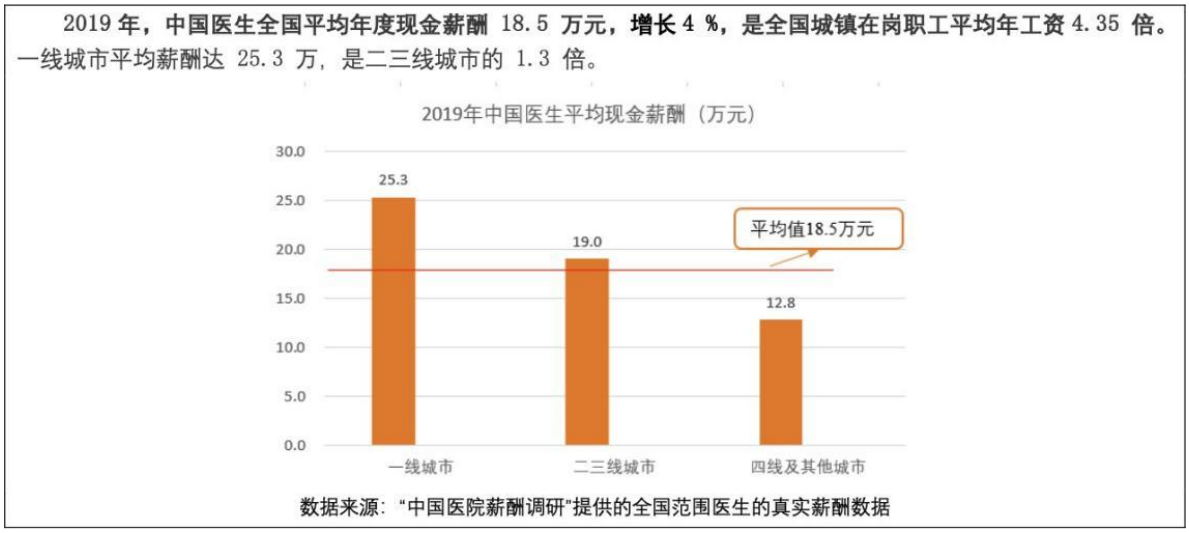
Since COVID-19, doctors, along with nurses and other frontline healthcare workers, have become the "most beautiful counter-marchers" and true heroes—they risk their own lives to serve others! Who says that doctors are all frail scholars? They vow not to return until the pandemic is defeated! Thousands of doctors across the country have demonstrated the responsibility of being a physician during this pandemic. As Academician Zhong Nanshan said, he believes they are all heroes!



The People's Daily once praised Zhong Nanshan as having both the responsibility of a national figure and the courage of a warrior. The phrase he has most often responded with is, "I am merely a doctor treating patients." Wearing this white coat is a choice to give one's all. Inspired by Zhong Nanshan, over 10,000 young medical workers nationwide have already gone to Hubei. They are dedicated and meticulous in their duties, simply because of their benevolent hearts and their commitment to not failing their mission as healers.

Treatment 4. Salary information

Original version



English translation

In 2019, the average annual cash salary of doctors in China was RMB 185,000, an increase of 4% and 4.35 times the average annual salary of urban employees nationwide. The average salary in first-tier cities was RMB 253,000, about 1.3 times the salary in second- and third-tier cities.

Average annual salary of doctors in China in 2019

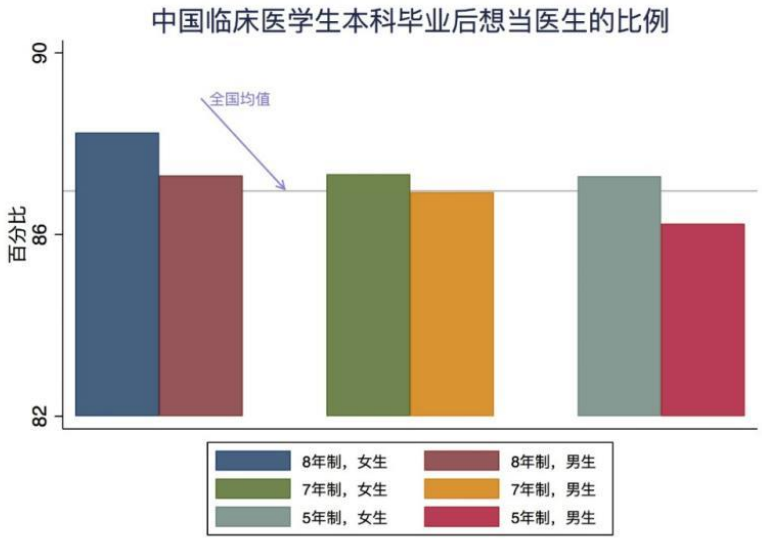
City category	Annual salary (RMB)
First-tier cities	253,000
Second- and third-tier cities	190,000
Other cities	128,000
National average	185,000

Data source: China Hospital Salary Survey. The English translation presents the values shown in the original chart.

Treatment 5. Peer career intentions

Original version

根据 2019 年全国临床医学本科毕业生调查，87%的中国临床医学生本科毕业后想继续做医生；女生的比例高于男生，长学制的学生略高于 5 年制学生。



English translation

According to the China Medical Student Survey (CMSS) 2019, 87% of medical students in China want to continue to be doctors after undergraduate graduation. This proportion is higher among female students than male students, and it is slightly higher among students in longer programs compared to those in 5-year programs.

